

WaiYan(Anson) Chan

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Education

University of Wisconsin-Madison

Madison, WI

Ph.D. in Electrical and Computer Engineering

Present

M.S. in Electrical and Computer Engineering

June 2024

B.S. in Mechanical Engineering

December 2021

Relevant Coursework

- Dynamics and Control of AC Drives
- Redesign and Prototype Fabrication
- Automatic Controls Lab
- EM Design of AC Machines
- Power Electronic Circuits and Lab
- Finite Elements

Research Experience

Precision Mechatronics and Control Lab - WEMPEC

Aug. 2024 – Present

Graduate Research Assistant, Ph.D. - *Advisor: Lei Zhou*

Madison, WI

Severson Research Group - WEMPEC

Aug. 2022 – Jun. 2024

Graduate Research Assistant, M.S. - *Advisor: Eric L. Severson*

Madison, WI

- **Additively Manufactured (AM) Stator Housing for High-Speed Bearingless Generator**
 - Developed and fabricated AM housing with integrated cooling channels for a 100kW, 80kRPM high-speed twin bearingless generator in a microturbine-based CHP generation system.
- **Bearingless Machines (BSPM) for Aerial E-Turbocharger Application**
 - Conducted structural and modal analysis to validate rotor design by identifying critical speed and rotor stress.
 - Modeled and conducted FE analysis of an 8kW 4-DOF twin-stator BSPM, along with component fabrication.
 - Facilitated the development of multi-physics modeling framework for BSPM and created Python scripts for evaluating machine constants and coil inductances in JMAG.

Wisconsin Electric Machinery and Power Electronics Consortium

Sep. 2019 – Dec. 2021

Undergraduate Researcher

Madison, WI

- Prototyped a desktop size dynamometer setup capable of characterizing electric machines rated up to 10 N-m of torque and 3000 RPM. Reduced the original setup volume by 35%.
- Collaborated with a team of graduate researchers to prototype a modular terminal box for powering a 200 kW NASA prototype machine. Responsible for sourcing parts and creating CAD models.
- Fabricated BSPM machines by producing stator slot windings and machining components used in the assembly.

Industry Experience

Milwaukee Tool

Jan. – Aug. 2022

NPD Mechanical Design Engineer

Brookfield, WI

- Conducted FE analysis on BLDC machines of various sizes and winding configurations, recommended optimal designs for product development using Pugh Matrix for informed and cost-effective solutions.
- Assessed power tool performance requirements through the collection and analysis of motor thermal characteristics and power output data from competitor products.

Projects & Skills

eMach | *Open-Source Python Machine Modeling and Optimization Framework*

2022 - Present

UW CoE Undergraduate Learning Center | *Tutor, Physics and Engineering Statics*

2018-19

Wisconsin Racing Electric | *Chassis Team Member*

2018-21

Hands-on: Electric Machine Fabrication, Component Machining and Assembly, CAD Design and Drafting, GD&T

Tools: Python, Git, SolidWorks, NX, ANSYS, MATLAB / Simulink, Altium, LabView, MAGNET, FEMM, JMAG, $\text{\LaTeX}$

Publications

1. T. Noguchi, N. Petersen, **W. Chan**, L. Rapp, E. Severson, "Bearingless Motor/Generator Applications in sCO₂ Power Cycles," *The 8th International Supercritical CO₂ Power Cycles Symposium, San Antonio, TX, USA, 2024 (Accepted)*
2. T. S. Slininger, **W. Chan**, E. L. Severson, and B. Jawdat, "An Overview on Passive Magnetic Bearings," *2021 IEEE International Electric Machines & Drives Conference (IEMDC), Hartford, CT, USA, 2021*